

# TPT Agent

## Algorithm Documentation

### Tradier Paper Trading — Full Strategy Reference

|          |                                                      |
|----------|------------------------------------------------------|
| Strategy | Cash-Secured Puts (CSP) + LEAPS Deep ITM Calls       |
| Broker   | Tradier Sandbox (Paper Trading) — Account VA54665450 |
| Schedule | Daily at 6:35 AM PT (every day)                      |
| Data     | Tradier real-time APIs + yfinance (earnings only)    |
| Discord  | #beta-ai-trades — Summary cards only                 |
| Version  | Generated June 14, 2026                              |

# 1. Overview

The TPT Agent is a fully automated options trading bot that runs on Tradier's paper trading sandbox. It executes two parallel options strategies on the same curated stock watchlist: Cash-Secured Puts (CSP Wheel strategy) and LEAPS deep ITM calls (stock replacement). The bot runs once per day at 6:35 AM PT, screens the entire approved stock universe, manages existing positions, and opens new ones based on a rules-based scoring system.

**Key design principles:** (1) Never trade below the 200-day SMA. (2) Never sell a CSP into an earnings window. (3) Size up only on high-conviction setups (score 5/5). (4) Close winners early — CSPs at 50% premium captured, LEAPS at +5% profit. (5) VIX gates deployment amount; LEAPS gate is  $VIX \leq 18$  OR  $\geq 21$ .

## System Architecture

| Component        | Detail                                                           |
|------------------|------------------------------------------------------------------|
| Bot file         | tpt_agent.py                                                     |
| Scheduler        | macOS launchd — com.ankushsinghal.tptagent.plist                 |
| Trigger time     | 6:35 AM PT, every day                                            |
| Broker           | Tradier sandbox (paper trading)                                  |
| Account          | VA54665450                                                       |
| Stock universe   | Google Sheets CSV — curated approved watchlist                   |
| Options data     | Tradier markets/options/chains (greeks=true) — real-time         |
| Stock price data | Tradier markets/history + markets/quotes — real-time             |
| VIX              | Tradier markets/quotes (\$VIX.X) — yfinance fallback             |
| Earnings filter  | yfinance only (Tradier has no earnings calendar)                 |
| Order type       | Limit orders at live mid-price (bid+ask)/2                       |
| Log file         | tpt_agent.log                                                    |
| Discord channel  | #beta-ai-trades (hardcoded webhook — no other channel)           |
| Discord output   | Run summary + open positions table + CSP summary + LEAPS summary |

## 2. Data Sources

All market data is fetched in real-time via Tradier's REST APIs. yfinance is retained only for earnings dates, which Tradier does not provide.

| Data Point                  | Source                               | Latency   | Used For                                                        |
|-----------------------------|--------------------------------------|-----------|-----------------------------------------------------------------|
| Stock OHLCV bars            | Tradier /markets/history             | Real-time | SMA 20/50/200, Bollinger Bands, RSI                             |
| Real-time stock quote       | Tradier /markets/quotes              | Real-time | Current price, day change %                                     |
| Options chain (bid/ask/OI)  | Tradier /markets/options/chains      | Real-time | Strike selection, ARR, OI filter                                |
| Implied Volatility (IV)     | Tradier greeks.mid_iv                | Real-time | IV scoring criterion, BS delta fallback                         |
| Options Delta (CSP)         | Tradier greeks.delta                 | Real-time | Delta filter vs BB-position-based range (per stock)             |
| Options Delta (LEAPS)       | Black-Scholes computed               | Real-time | Always recomputed — Tradier greeks unreliable for 1-2yr options |
| VIX                         | Tradier \$VIX.X → yfinance fallback  | Real-time | Deploy % and LEAPS gate                                         |
| Earnings dates              | yfinance (only yfinance use)         | Daily     | Hard filter — skip if earnings in DTE window                    |
| Option expirations          | Tradier /markets/options/expirations | Real-time | Finding valid DTE windows                                       |
| Live mid-price at execution | Tradier /markets/quotes              | Real-time | Limit price on every order placed                               |

### 3. Run Pipeline — 9 Phases

Every time the bot fires at 6:35 AM PT it executes nine sequential phases. Each phase depends on the results of the previous one.

**1**

#### VIX + Account Snapshot

Fetch VIX, portfolio value, cash, and all open positions from Tradier.

**2**

#### Manage Existing Positions

Profit exits placed automatically. Loss exits post Discord alert — trader closes manually.

**3**

#### Refresh Capital After Closes

Re-fetch account after any closes to get updated cash and buying power.

**4**

#### Load Tickers + Hard Filters

Pull approved stock list, fetch price history, apply 200 SMA and earnings hard filters.

**5**

#### CSP Screening + Scoring

Score each stock on 5 technical signals, find best put contract, rank by score then ARR/Delta ratio.

**6**

#### LEAPS Screening

If  $VIX \leq 18$  or  $\geq 21$ : score stocks on 3 LEAPS criteria, find best deep ITM call contract.

**7**

#### Execute LEAPS Trades

Top-ranked first, 1 contract per pick, until 15% portfolio budget exhausted.

**8**

#### Execute CSP Trades

Top-scored first, 1 contract per pick, until VIX-adjusted cash budget exhausted.

**9**

#### Post Discord Summary

Post run summary, open positions table, CSP summary table, LEAPS summary table.

## 4. Phase 1 — VIX & Account State

### CSP Delta — BB-Based Per Stock (replaces VIX-global range)

CSP put delta is no longer determined by VIX globally. Each stock gets its own delta range based on its Bollinger Band position at screening time. See Phase 5 (Section 7) for the full 3-tier table.

### VIX → Cash Deployment Percentage

| VIX Level | Deploy % | Rationale                                      |
|-----------|----------|------------------------------------------------|
| > 25      | 80%      | Fear = opportunity, deploy aggressively        |
| 20 – 25   | 70%      | Elevated vol, still good entry conditions      |
| 15 – 20   | 60%      | Moderate vol, standard deployment              |
| 12 – 15   | 40%      | Low vol, be selective and conservative         |
| < 12      | 20%      | Very low vol, premiums too thin to deploy much |

### Budget Calculations

- **CSP Budget** = Cash × VIX deploy %
- **LEAPS Budget** = Portfolio Value × 15% – Current LEAPS Exposure (never exceeds 15% of portfolio total)
- **LEAPS VIX Gate** = Screen and execute LEAPS when **VIX ≤ 18 OR VIX ≥ 21**

*Rationale for LEAPS VIX gate (≤18 or ≥21):* VIX ≤ 18 = calm market, stocks fairly valued — ideal LEAPS entry. VIX ≥ 21 = fear-driven selloff, stocks oversold — excellent LEAPS entry at depressed prices. Zone 18–21 excluded: moderate stress where IV is neither cheap nor justified.

## 5. Phase 2 — Position Management

Before opening any new positions, the bot scans every open position. Three independent rules evaluate each position. **Profit exits are placed automatically. Loss exits post a Discord alert — the trader decides manually.**

- **Auto-order:** Profit exits only — bot places the closing order immediately.
- **Discord alert:** Loss exits only — bot posts a recommendation, no order placed. Trader reviews and closes manually.

### CSP Exit Rules — Short Puts (checked in priority order)

| Rule                              | Trigger                                                                        | P&L; State    | Action                  |
|-----------------------------------|--------------------------------------------------------------------------------|---------------|-------------------------|
| #1 — 50% Capture                  | $(\text{Avg Entry} - \text{Current}) / \text{Avg Entry} \geq 50\%$             | Profit        | ■ AUTO — BUY TO CLOSE   |
| #2 — 21-DTE Exit (flat or profit) | $\text{DTE} \leq 21$ AND $\text{captured} \geq 0\%$                            | Profit / Flat | ■ AUTO — BUY TO CLOSE   |
| #2 — 21-DTE Exit (small loss)     | $\text{DTE} \leq 21$ AND $-50\% \leq \text{captured} < 0\%$                    | Small loss    | ■■ ALERT — manual close |
| #3 — Stop-Loss                    | $\text{Current} \geq \text{Avg Entry} \times 3.0$ (loss = 2× premium received) | Loss          | ■■ ALERT — manual close |

### LEAPS Exit Rules — Long Calls

| Rule               | Trigger                                                             | P&L; State | Action                  |
|--------------------|---------------------------------------------------------------------|------------|-------------------------|
| #1 — Profit Target | $(\text{Current} - \text{Avg Entry}) / \text{Avg Entry} \geq 5\%$   | Profit     | ■ AUTO — SELL TO CLOSE  |
| #2 — Stop-Loss     | $(\text{Current} - \text{Avg Entry}) / \text{Avg Entry} \leq -50\%$ | Loss       | ■■ ALERT — manual close |

### Discord Loss Alert Format

When a loss exit triggers, the bot posts an alert to **#beta-ai-trades** containing:

- Position (compact format: e.g. NVDA \$195P Jun26)
- Entry price → current price
- Unrealized P&L (\$ and %)
- Trigger reason (which rule fired)
- Suggested action (BUY/SELL TO CLOSE N contracts at market)
- Confirmation: *No order placed — awaiting your decision.*

### Order Execution Mechanics (profit exits only)

- Fetch live bid/ask from Tradier **markets/quotes** endpoint
- Calculate mid-price:  $(\text{bid} + \text{ask}) / 2$
- Place a **limit order at mid-price**, time-in-force = day
- If no live quote available: fallback to last known price

## 6. Phase 4 — Stock Universe & Hard Filters

The bot loads a curated watchlist from Google Sheets, fetches 310 calendar days of daily OHLCV bars from Tradier for each ticker, computes technical indicators, then applies two hard filters. Any stock failing either filter is completely excluded from both CSP and LEAPS screening.

### Technical Indicators Computed

| Indicator       | Calculation                                                  | Used In                                                                                    |
|-----------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| SMA 20          | Mean of last 20 daily closes                                 | Bollinger Band midline, CSP scoring                                                        |
| SMA 50          | Mean of last 50 daily closes                                 | CSP scoring (strike above 50 SMA)                                                          |
| SMA 200         | Mean of last 200 daily closes                                | Hard filter — must be above this                                                           |
| Bollinger Upper | $SMA\ 20 + 2 \times std(\text{last 20 closes})$              | Reference                                                                                  |
| Bollinger Lower | $SMA\ 20 - 2 \times std(\text{last 20 closes})$              | LEAPS scoring (near lower band)                                                            |
| RSI (14)        | Wilder's smoothed 14-period RSI                              | CSP hard gate ( $\geq 65$ =skip) + CSP scoring ( $< 50$ =+1) + LEAPS scoring ( $< 70$ =+1) |
| Day Change %    | $(\text{current} - \text{prev\_close}) / \text{prev\_close}$ | CSP scoring (healthy pullback)                                                             |

### Hard Filters

| Filter                    | Logic                          | Skip if                                                 |
|---------------------------|--------------------------------|---------------------------------------------------------|
| 200-day SMA               | Is stock in long-term uptrend? | Price $\leq$ SMA 200                                    |
| Earnings in DTE window    | Earnings report coming up?     | Earnings date falls within next MAX_DTE (45) days       |
| RSI overbought (CSP only) | Applied in Phase 5 per stock   | RSI $\geq 65$ — stock is overbought, skip CSP screening |

■ Earnings data is fetched from yfinance — the only remaining yfinance dependency. Tradier does not provide an earnings calendar.

## 7. Phase 5 — CSP Screening & Scoring

### Step 1 — RSI Hard Gate

If  $RSI \geq 65 \rightarrow$  **SKIP immediately**. Overbought stocks have high mean-reversion risk within the 30–45 DTE window. No options lookup is performed.

### Step 2 — Pre-Score Gate (skip chain fetch early)

Before fetching any option chain, compute the **stock-level pre-score** (max 3): price below BB midline, healthy pullback today,  $RSI < 50$ . The other 2 score points (strike  $> 50$  SMA,  $IV \geq 40\%$ ) are contract-level. Since  $MIN\_SCORE\_TO\_TRADE = 3$  and at most 2 contract points can be added, any stock with pre-score  $< 1$  can never qualify  $\rightarrow$  **skip the chain fetch**. Correctness-preserving: it cannot drop a stock that would have qualified.

### Step 3 — BB-Based Delta Range

| BB Position                  | Delta Range                | Rationale                                                                               |
|------------------------------|----------------------------|-----------------------------------------------------------------------------------------|
| Price $\leq$ Lower BB + 3%   | 0.25 – 0.35 (Aggressive)   | Stock oversold $\rightarrow$ high reversal probability $\rightarrow$ sell closer to ATM |
| Between Lower BB and Mid BB  | 0.15 – 0.25 (Moderate)     | Partial pullback $\rightarrow$ standard entry quality                                   |
| Price $\geq$ Mid BB (SMA 20) | 0.08 – 0.15 (Conservative) | At/above average $\rightarrow$ more downside room $\rightarrow$ stay far OTM            |

### Step 4 — Technical Scoring (max 5 points)

| Signal                 | Points | Condition                                                        |
|------------------------|--------|------------------------------------------------------------------|
| Above 50-day SMA       | +1     | Stock price $>$ SMA 50 (medium-term uptrend)                     |
| At/below 20-day SMA    | +1     | Stock price $\leq$ SMA 20 / BB midline (short-term pullback)     |
| Healthy pullback today | +1     | Day change between $-0.5\%$ and $-5.0\%$                         |
| $RSI < 50$             | +1     | Actively oversold — best entry timing (replaces price $<$ \$100) |
| $IV \geq 40\%$         | +1     | Elevated implied volatility — premium is rich                    |

Require final score  $\geq MIN\_SCORE\_TO\_TRADE$  (3) to qualify. Score = 5  $\rightarrow$  flagged HIGH CONVICTION.

### Step 5 — Find Best Put Contract (Tradier Options Chain)

For each expiration date between  $MIN\_DTE$  (30) and  $MAX\_DTE$  (45) days:

- Fetch full options chain from Tradier with **greeks=true**
- Filter puts: Open Interest  $\geq 50$  (rejects OI = 0), valid bid/ask quote
- Get delta from **Tradier greeks.delta**; fallback to Black-Scholes if null
- Apply BB-position-based delta filter (per-stock tier: aggressive / moderate / conservative)
- Calculate **ARR** = (mid / strike)  $\times$  (365 / DTE)  $\times$  100

**Strike Selection Logic (ARR band 40–70%):**



**Contracts within band ( $40\% \leq \text{ARR} \leq 70\%$ ):** Pick the highest ARR in band.

**If ALL contracts are above the cap ( $\text{ARR} > 70\%$ ):** Step down to the lowest (safest) ARR above the cap.

**If NO contract reaches the floor (all  $\text{ARR} < 40\%$ ):** Skip the stock entirely — MIN\_ARR is a HARD floor, never breached. (This is why e.g. a 30%-ARR contract will not be selected.)

## Step 6 — Rank and Select Top 5

- Sort qualifying CSPs by the following priority:
- **1. Score DESC** — technical setup quality (primary)
- **2.  $\text{ARR} \div \text{Delta}$  DESC** — normalized risk-adjusted return (secondary)
- Rationale: a lower delta + higher ARR means the market pays you MORE for taking LESS
- directional risk. ARR/Delta correctly ranks e.g. delta=0.14 ARR=69% (ratio=493)
- above delta=0.20 ARR=47% (ratio=235). Pure ARR alone would invert this.
- **3. DTE DESC** — more time = more cushion for recovery
- **4. ARR DESC** — final tie-breaker only
- Take top 5

## 8. Phase 6 — LEAPS Screening

**VIX Hard Gate:** LEAPS screening runs ONLY when  $VIX \leq 18$  OR  $VIX \geq 21$ . Zone 18–21 is excluded (moderate stress — IV inflated without directional clarity). Outside both zones the entire LEAPS phase is skipped.

Runs on the same stock universe that passed the hard filters in Phase 4.

### LEAPS Scoring Criteria (3 points total — all stock-level)

| Criterion          | Points | Logic           | Notes                                         |
|--------------------|--------|-----------------|-----------------------------------------------|
| Above 200-day SMA  | +1     | Price > SMA 200 | Long-term uptrend confirmed                   |
| Above 50-day SMA   | +1     | Price > SMA 50  | Medium-term momentum intact                   |
| RSI not overbought | +1     | RSI(14) < 70    | Wilder's smoothed RSI from Tradier daily bars |

- Minimum score to qualify for LEAPS: **2 out of 3**
- All three criteria are **stock-level** — the score is computed BEFORE any option chain is fetched, so chains are only pulled for stocks that already qualify.

### LEAPS Contract Selection (Tradier Options Chain)

- Expiration window: **365 to 730 days (1–2 years out)**
- **Farthest expiry first:** fetch the longest-dated expiry in the window (max time value, lowest theta — ideal for stock replacement). Fall back to the next-farthest only if it yields no qualifying contract. ~1 chain/stock vs 6–10 before.
- Contract type: **Call options only**
- OI filter: skip only if confirmed OI > 0 but < 50 (OI = 0 means data unavailable, allow through)
- Price: (bid+ask)/2 → ask → bid → lastPrice (in priority order)
- Delta: use **Tradier greeks.delta** if available, otherwise compute via **Black-Scholes**
- For BS computation: use Tradier IV if  $IV \geq 0.20$ , else default to **0.45** (LEAPS greeks are unreliable in data feeds)
- Delta filter:  **$0.70 \leq \text{delta} \leq 0.85$**  (deep in the money)
- Selection: within the chosen expiry, pick the contract whose delta is **closest to 0.77 (target)**

### LEAPS Ranking

- Sort by: **leaps\_score DESC**, then **RSI ASC** (most oversold first)
- Take top 5

## 9. Phases 7 & 8 — Trade Execution

### LEAPS Execution (Phase 7)

**Budget:** Portfolio Value × 15% (minus current LEAPS exposure already on the books)

**Allocation:** Top-ranked first. 1 contract per pick. Full remaining budget available for each pick.  
Skip a pick only if the remaining budget is too small for its cost — move to the next pick.

**For each LEAPS pick (in ranked order):**

- If remaining budget < cost\_per\_contract → skip, try next
- Fetch live mid-price from Tradier quotes
- Place **limit BUY TO OPEN** order at mid-price, qty = 1, time-in-force = day
- Deduct cost from remaining budget and continue to next pick

### CSP Execution (Phase 8)

**Budget:** Cash × VIX deploy %

**Collateral:** Each CSP requires Strike × \$100 in cash as collateral.

**Allocation:** Top-scored first. 1 contract per pick.

**For each CSP pick (in score-ranked order):**

- If remaining cash < strike × \$100 → skip, try next
- Fetch live mid-price from Tradier quotes
- Place **limit SELL TO OPEN** order at mid-price, qty = 1, time-in-force = day
- Deduct collateral from remaining cash and continue to next pick

### Order Format (Tradier API)

All orders are placed as form-encoded POST requests to Tradier:

| Parameter     | CSP Value         | LEAPS Value       |
|---------------|-------------------|-------------------|
| class         | option            | option            |
| symbol        | Underlying ticker | Underlying ticker |
| option_symbol | OCC put symbol    | OCC call symbol   |
| side          | sell_to_open      | buy_to_open       |
| quantity      | 1                 | 1                 |
| type          | limit             | limit             |
| price         | Live Tradier mid  | Live Tradier mid  |
| duration      | day               | day               |

## 10. Key Parameters Reference

All parameters are set via environment variables in the launchd plist (com.ankushsinghal.tptagent.plist). The defaults below are the current live values.

### CSP Parameters

| Parameter                    | Default   | Description                                                                     |
|------------------------------|-----------|---------------------------------------------------------------------------------|
| MIN_DTE                      | 30        | Minimum days to expiration for CSP puts                                         |
| MAX_DTE                      | 45        | Maximum days to expiration for CSP puts                                         |
| MIN_ARR                      | 40%       | Minimum annualized return on risk to qualify                                    |
| MAX_ARR                      | 70%       | ARR cap — step down to safer strike if exceeded                                 |
| MIN_OPEN_INTEREST            | 50        | Minimum open interest for liquidity (rejects OI = 0 — CSP needs real liquidity) |
| CSP_RSI_OVERBOUGHT           | 65        | Hard gate — skip stock if RSI $\geq$ this (overbought)                          |
| CSP_DELTA_NEAR_LOWER_MIN/MAX | 0.25/0.35 | Delta range when price $\leq$ lower BB + 3% (aggressive)                        |
| CSP_DELTA_MID_ZONE_MIN/MAX   | 0.15/0.25 | Delta range when price between lower and mid BB (moderate)                      |
| CSP_DELTA_ABOVE_MID_MIN/MAX  | 0.08/0.15 | Delta range when price $\geq$ mid BB (conservative)                             |
| MIN_SCORE_TO_TRADE           | 3         | Minimum score (out of 5) to qualify for execution                               |
| SIZE_UP_SCORE                | 5         | Score threshold for HIGH CONVICTION label                                       |
| SCORE_IV_MIN_PCT             | 40%       | IV threshold for +1 scoring point                                               |
| SCORE_RSI_OVERSOLD           | 50        | RSI below this earns +1 score point (actively oversold)                         |
| SCORE_DOWN_TODAY_MIN_PCT     | 0.5%      | Min pullback today to earn the pullback score point                             |
| SCORE_DOWN_TODAY_MAX_PCT     | 5.0%      | Max pullback today (beyond this = potential problem)                            |
| CSP_CLOSE_PREMIUM_PCT        | 50%       | Close CSP when this % of premium has been captured                              |
| CSP_DTE_EXIT                 | 21        | Close CSP at DTE $\leq$ this with positive/neutral P&L; (gamma gate)            |
| CSP_STOP_LOSS_MULT           | 2.0×      | Close CSP when loss $\geq 2\times$ premium received                             |

### LEAPS Parameters

| Parameter          | Default  | Description                                                      |
|--------------------|----------|------------------------------------------------------------------|
| LEAPS_MIN_DTE      | 365 days | Minimum DTE for LEAPS contracts                                  |
| LEAPS_MAX_DTE      | 730 days | Maximum DTE for LEAPS contracts (1–2 years)                      |
| LEAPS_MIN_DELTA    | 0.70     | Minimum call delta (lowered from 0.80)                           |
| LEAPS_MAX_DELTA    | 0.85     | Maximum call delta (lowered from 0.99)                           |
| LEAPS_TARGET_DELTA | 0.77     | Target delta — pick contract closest to this (lowered from 0.85) |
| LEAPS_VIX_CALM_MAX | 18       | LEAPS enabled when VIX $\leq$ this (calm zone)                   |
| LEAPS_VIX_FEAR_MIN | 21       | LEAPS also enabled when VIX $\geq$ this (fear/opportunity zone)  |

| Parameter               | Default | Description                                                          |
|-------------------------|---------|----------------------------------------------------------------------|
| LEAPS_MIN_SCORE         | 2       | Minimum LEAPS criteria score (out of 3)                              |
| LEAPS_MIN_OI            | 50      | Minimum OI (only enforced when OI > 0 from API)                      |
| LEAPS_MAX_PORTFOLIO_PCT | 15%     | Hard cap on total LEAPS exposure as % of portfolio (raised from 10%) |
| LEAPS_RSI_OVERBOUGHT    | 70      | RSI threshold — below this = not overbought (+1 pt)                  |
| LEAPS_RSI_PERIOD        | 14      | RSI calculation period (Wilder's smoothed)                           |
| LEAPS_CLOSE_PROFIT_PCT  | 5%      | Close LEAPS when unrealized profit reaches this (lowered from 25%)   |
| LEAPS_STOP_LOSS_PCT     | 50%     | Close LEAPS when unrealized loss reaches this                        |

## Performance Parameters

| Parameter   | Default | Description                                                         |
|-------------|---------|---------------------------------------------------------------------|
| MAX_WORKERS | 6       | Thread-pool size for parallel per-ticker screening (Phases 4, 5, 6) |

*Performance: per-ticker screening runs in parallel; option expirations are cached per ticker; LEAPS fetch only the farthest expiry; CSP and LEAPS gate on stock-level score before any chain fetch. A full run takes  $\approx 1$  minute (down from  $\approx 7$  minutes).*

## 11. Discord Output

All output goes exclusively to **#beta-ai-trades**. The webhook URL is hardcoded in `tpt_agent.py` — it cannot be redirected to another channel via configuration. Only summary tables are posted; no individual trade detail cards.

| Post # | Content                   | Always Posted?                | Format                                                                                 |
|--------|---------------------------|-------------------------------|----------------------------------------------------------------------------------------|
| 1      | Run Summary               | Yes                           | Embed: balance, VIX, profit closes (auto-ordered), loss alerts (manual), new positions |
| 2      | Open Positions Table      | If positions exist            | Code block: position, qty, P&L; \$, P&L; %                                             |
| 3      | Loss Alert (per position) | If loss exit triggered        | Red embed: position, P&L;, trigger reason, suggested action. No order placed.          |
| 4      | LEAPS Criteria Embed      | Yes                           | Embed: VIX gate status, criteria list, DTE/delta targets                               |
| 5      | LEAPS Summary Table       | If VIX in range + picks found | Code block: ticker, strike, delta, expiry, cost                                        |
| 6      | CSP Summary Table         | If picks found                | Code block: ticker, strike, delta, expiry, premium, ARR                                |

All embeds and tables include `[TPT]` label to distinguish from the Experiment Bot which posts to the same channel.

## 12. Full Decision Flow

START 6:35 AM PT trigger

Fetch VIX → determine deploy %; LEAPS gate:  $VIX \leq 18$  OR  $\geq 21$

Fetch Tradier account: portfolio value, cash, option buying power

Fetch all open positions

### ■■■ POSITION MANAGEMENT ■■■

For each open SHORT PUT (CSP):

- If stop-loss triggered ( $\text{loss} \geq 2 \times \text{premium}$ ) → ■■ DISCORD ALERT only, no order
- If 50% premium captured → ■ BUY TO CLOSE at mid (auto)
- If  $\text{DTE} \leq 21$  and in profit → ■ BUY TO CLOSE at mid (auto)
- If  $\text{DTE} \leq 21$  and at loss → ■■ DISCORD ALERT only, no order

For each open LONG CALL (LEAPS):

- If stop-loss triggered ( $\text{loss} \geq 50\%$ ) → ■■ DISCORD ALERT only, no order
- If profit target reached ( $\text{gain} \geq 5\%$ ) → ■ SELL TO CLOSE at mid (auto)

Re-fetch account to capture released capital

### ■■■ STOCK SCREENING ■■■

Load approved ticker list from Google Sheets

For each ticker (parallel, ThreadPoolExecutor —  $\text{MAX\_WORKERS}=6$ ):

- Fetch 310 days OHLCV + quote from Tradier → compute SMA20/50/200, BB, RSI
- HARD FILTER 1: skip if  $\text{price} \leq \text{SMA } 200$
- HARD FILTER 2: skip if earnings within  $\text{MAX\_DTE}$  days (yfinance)

### ■■■ CSP SCORING (parallel) ■■■

For each stock passing hard filters:

- Hard gate:  $\text{RSI} \geq 65$  → skip (overbought)
- Pre-score gate: skip chain fetch if stock-level pre-score can't reach 3
- BB-based delta range: near lower  $\text{BB}=0.25\text{--}0.35$  / between  $\text{BBs}=0.15\text{--}0.25$  / above mid= $0.08\text{--}0.15$
- Fetch put chain from Tradier (greeks=true)
- Filter puts:  $\text{OI} \geq 50$  (rejects  $\text{OI}=0$ ), delta in BB-tier range, ARR 40–70%
- Score: strike  $> 50$  SMA (+1), below mid BB (+1), pullback 0.5–5% (+1),  $\text{RSI} < 50$  (+1),  $\text{IV} \geq 40\%$  (+1)
- If final score  $< 3$ : skip

Sort CSPs: score DESC → ARR/Delta DESC → DTE DESC → ARR DESC → take top 5

### ■■■ LEAPS SCORING (parallel; only if $VIX \leq 18$ OR $\geq 21$ ) ■■■

For each stock passing hard filters:

- Score FIRST (stock-level): above 200 SMA (+1), above 50 SMA (+1),  $\text{RSI} < 70$  (+1)
- If score  $< 2$ : skip (no chain fetch)
- Fetch ONLY the farthest expiry chain (DTE 365–730); fall back if no contract
- Filter calls:  $\text{OI} \geq 50$ , delta 0.70–0.85; pick closest to delta 0.77

Sort LEAPS by score DESC, RSI ASC → take top 5

### ■■■ EXECUTION ■■■

LEAPS: iterate top-ranked first, 1 contract each, until 15% budget exhausted

CSP: iterate top-scored first, 1 contract each, until VIX-deploy cash exhausted

All orders: limit at live Tradier mid-price, time-in-force = day

■■■ DISCORD ■■■

Post run summary, open positions, CSP table, LEAPS table to #beta-ai-trades

END

*\* The experiment\_bot.py (Alpaca) runs an identical strategy independently. Trade picks are generated separately for each broker — they are not shared. This allows side-by-side comparison of broker execution quality.*